



Workshop Report

UvA Master Programme Conservation and Restoration of Cultural Heritage

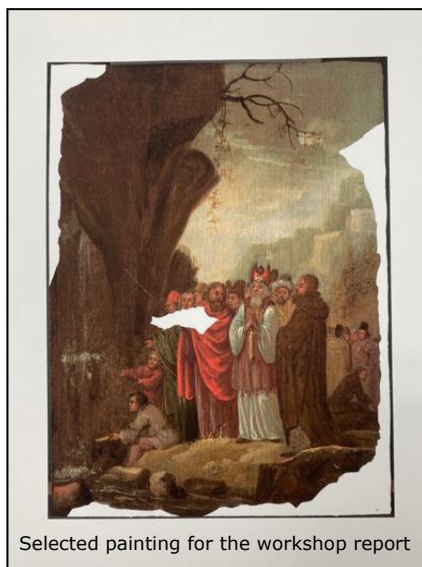
- Name of student (student number): Moje Ryoo (13837486)
- Specialisation: Contemporary Art
- Course: **Skill Building for Conservation 4**
Loss compensation and visual reintegration
- Submission Date: 25th of October 2024

*All the photos were taken by the student unless stated otherwise.

Retouching Techniques

Activity log

- **Date/ location:** 14th. Oct. 2024 (09:00~17:00) / BSH F2.01B-F2.01C
- **Supervisor:** Emilie Froment(PA), Mandy Slager(GC), Gwen Tauber(Guest lecturer)
- **Aim**
 - 1) Understanding each retouching technique (neutral tone, Tratteggio, chromatic selection, chromatic abstraction, pointillism, and imitative retouching) and applying different techniques to the test painting printed on the paper.
 - 2) Analysing the impact of the selected retouching techniques for the painting, use of pigments, hue, values, and brightness.
 - 3) Group discussions for the evaluation of the application techniques, limitations, and the final result to expand insights on the different approaches.
- **Material preparation:** A4 printed painting (damaged/ losses), watercolors, small sable brushes, pot for water, tissues, palette.

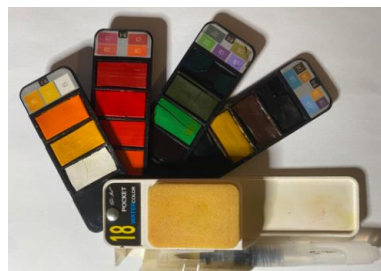


Oil painting

Oil painting

Terracotta

Tiles



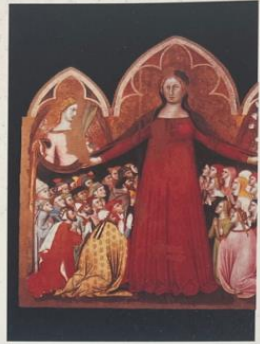
*The depth of the loss is not considered because the painting is printed on paper.

*The losses are shown as white areas as if the gap had been filled with a certain filler.



Methodology and Process record

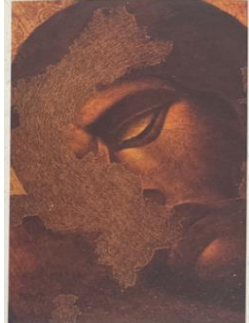
- Applicable retouching techniques ¹

Technique	Features and descriptions	Example image
Neutral tone	It is often used when neither the form nor detail can be reliably reconstructed. The retouching is applied in a uniform color to achieve a neutral appearance, without any attempt to restore the appearance of missing parts. The goal is to push the lacuna to the background and reduce to the minimum the risk of retouching due to subjective interpretations. The tone can be similar for the whole painting and may vary in lightness and hue per areas.	 2 Fig. 71 'Neutral tone' retouching by the restorer Tintori in 1978 on Bartolo di Fredi Cim's (active 1353-1410) <i>Madonna of Mercy</i>. Museo d'Arte Sacra, Pienza
Tratteggio (or Rigattino)	Tratteggio consists of the application of many vertical strokes of various colors that are gradually built up to approximate the appearance of the original from a normal viewing distance, while remaining discernible under closer examination. Depending on the assessment of the impact of the loss, tratteggio could be used to reconstruct the missing forms of the composition or as a general toning of the lost areas. The technique was intended to provide a precise formula that would mitigate subjective influences and personal idiosyncrasies in the execution of the retouching. (by AIC)	 3 Fig. 70 'Rigattino' retouching, Amico friuli di Desso, <i>Allegory</i>, c. 1518. Restoration completed in 1996 by Roberto Buda and Muriel Vervat. Uffizi, Florence
Chromatic selection (or Selezione)	The brushstrokes have to be homogeneous in intensity, length, size, distance, and brightness, neither too thin nor too thick. Dilution, length, and distance of the brushstrokes depends on the colour obtained, the distance at which the work will be read and the size of the work. Therefore, the strokes may vary in dimension and also in relation to the distance of reading. (by RCE brochure) *Additional information: There is no color	 4

¹Based on the workshop guide provided, which is extracted from various publications written by AIC, P. Philippot, Paolo, Laura Mora, Michel Eugène Chevreul, and RCE.

²Wallert, A p.50

³Wallert, A p.50

	mixing on the palette rather the visual illusion is achieved through the juxtaposition of 4 pure colors usually yellow/red/green/black. Green can be replaced by blue or green-blue and red by red-orange.	
Chromatic abstraction (or Astrazione Cromatica)	When the loss is significant and cannot be reintegrated without invention or without overpowering the remains of the original, then Astrazione Cromatica can be employed. In this case, the hatched brushstrokes do not reconstruct forms, and the color of the retouching is derived from the overall chromatic character of the painting. The abstraction is intended to create a “neutral link” between the passages of well-preserved original paint. (by AIC)	5  fig. 72 “Astrazione cromatica” retouching in detail of Cimabue’s Crucifixion of c. 1285, damaged in the flood of 1966, after restoration 1996 by Umberto Baldini and Ornella Casazza. Museo dell’Opera di Santa Croce, Florence
Pointillism	Juxtaposition of pure colored dots. The technique relies on the ability of the eye and mind of the viewer to blend the color spots into a fuller range of tones. Inspired by the color theory of Michel Eugène Chevreul (1786-1889), French chemist.	6 
Imitative retouching	Also known as mimetic or fully integrated retouching. The retouching seeks to match the losses to the surrounding areas as perfectly as possible. The inpainting should always remain distinguishable, although not necessarily in visible light. It may be necessary to use other diagnostic techniques, such as UV light to distinguish the inpainting from the original paint layer. (by RCE brochure)	7 

⁴<http://www.toscanarestauroarte.it/ricopitt.asp?idMenu=8&idLingua=2>

⁵Wallert, A p.50

⁶Raphaë “ Ange tenant un phylactère ” Musée_du_Louvre

⁷<https://artuk.org/discover/stories/sympathetic-retouching>

- Process record

- 1) Morning session:** applying two techniques, considering the reasons for the selection

- **Neutral tone:** Pushing the damage to the backward, making the audience focus on the original part of the painting. The relatively large area of damage. The original painting was painted on a wood panel, brown color can provide a natural visual effect as a ground material, combined with original paints.

- **Pointillism:** Not suggesting or inventing the original drawings and colors to the audience, making damages smudge to the surroundings. A relatively small area and damaged is located in the middle part of the painting, it must be reintegrated to a higher degree than the damaged parts on the edges

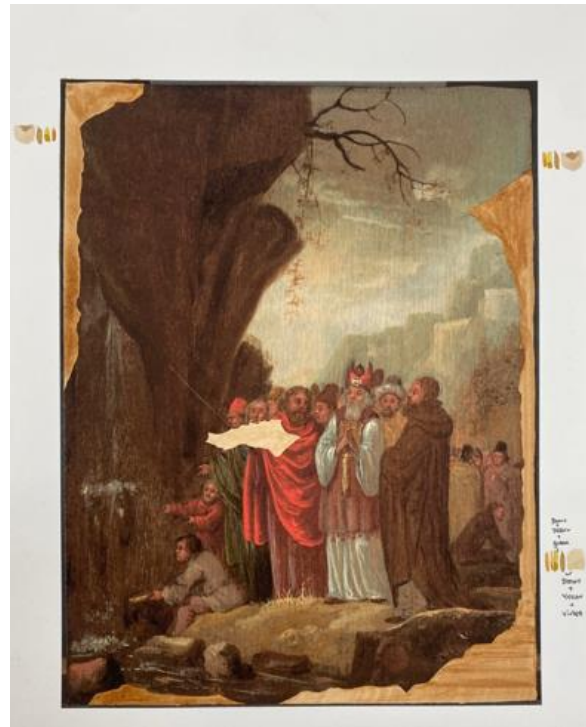
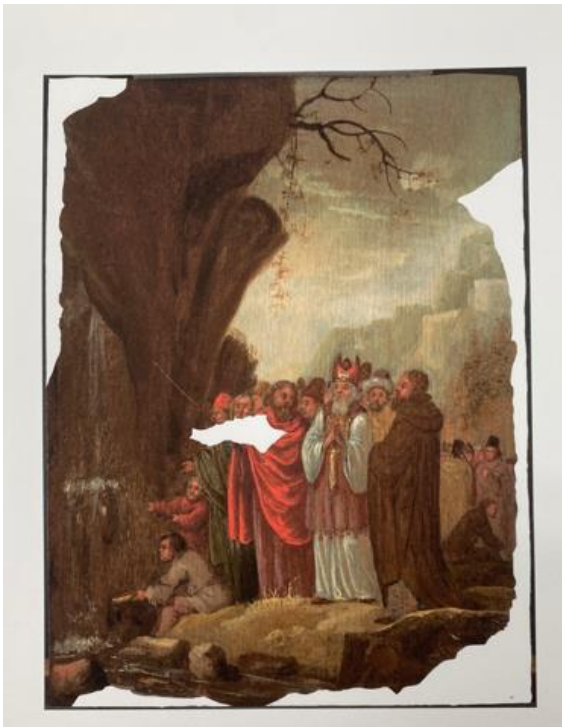
- 2) Afternoon session:** improvement of the retouching by mixed techniques

-Completing the retouching with **neutral tone, Tratteggio, pointillism, and imitative retouching** technique.

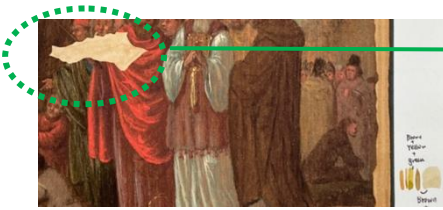
- Retouching practice details

- 1) Neutral tone**

- Before & After



- Mixing colors and matching colors



Brighter tone of brown color than the background. This area is located on the middle and has impact of the meaning, further integration is required.



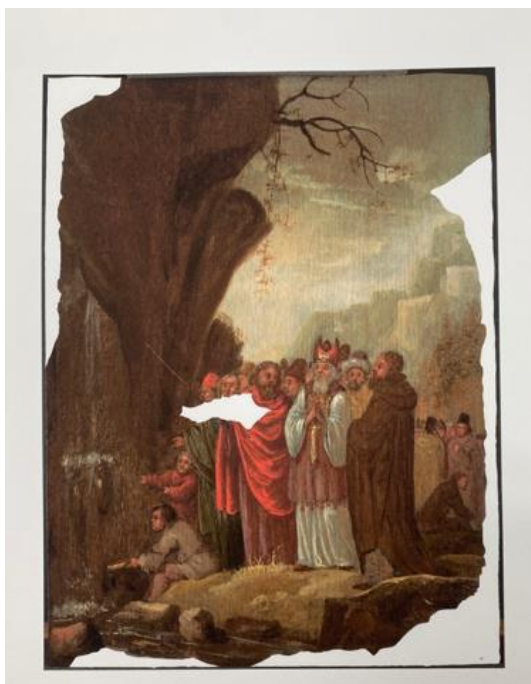
Brown + Yellow + Green color. Intuitively mixing the color, which is similar to surroundings in the original painting. Each part can have a different tone of color, but I choose only brown and differentiated the saturation.

* Notes

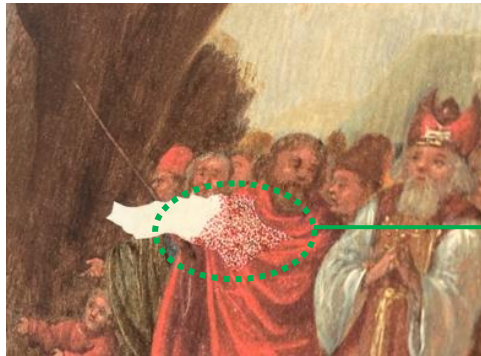
- Diluted watercolor paint made brush strokes, which is undesirable.
- Considering the paper, the texture will be different from the original painting
- The edges of the damage require more delicate treatment. If less retouched, there will be white filler showing, and also the conservator needs to avoid over-painting of damage as well.
- If the original painting has a brown color, applying a neutral tone is the recommended technique.

2) Pointillism

- Before & After



- Mixing colors for a middle area



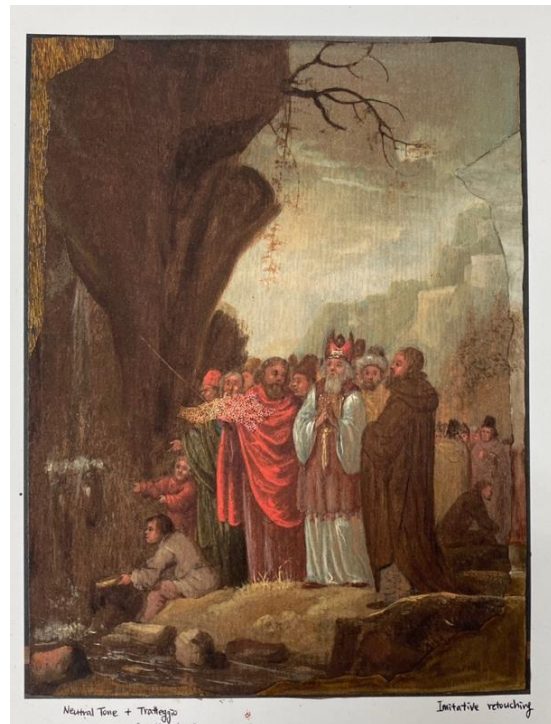
Brown + Orange + Red color.
Intuitively mixing the color, which is similar to the original color of the surroundings.
Pointillism helps the conservator to prevent inventing the drawing of the original paintings, presenting similar colors to compromise the interference of the visual.

* Notes

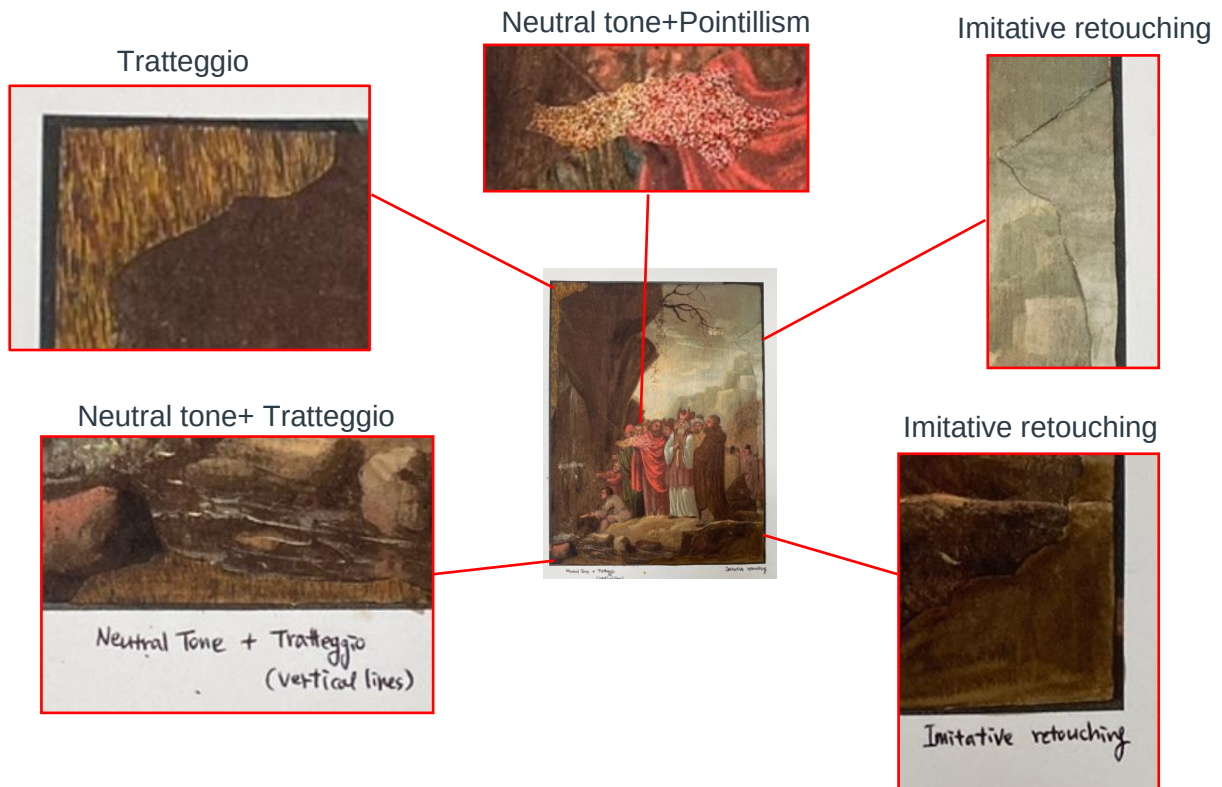
- On the background, over the white filler, the tone was down with the very light brown.
- Mini brush makes irregular dots, I have to change the tool to a pointy stick.
- Considering the scale, the pointillism will work better on the bigger (original) painting.
- Always check the wholeness of the painting while applying any techniques.
- There must not be blanks. (The ground color should not be shown.)

3) Mixed Technique: neutral tone, Tratteggio, pointillism, and imitative retouching

- Before & After



- Details of each techniques



-Process of Tratteggio

- 1) Considering the colors of surroundings
- 2) Brown + Orange + Red color
- 3) Practice strokes to check color match
- 4) Strokes: more and less 1cm, vertical
- 5) There must not be blanks



* Notes

- 1) Tratteggio requires proper brushes. It is very difficult to draw identical vertical lines.
- 2) The whole color tone will depend on the amount of lines drawn.

Guest lecture by Gwen Tauber, painting conservator at Rijks Museum, retired

- Recreating texture on the test panel was important as well as color-matching in retouching.
- After the removal of overpaints in the former treatment, many parts were revealed as loss.
- Historical context checking: the green knot on the cloth has to be a single knot.
Double(ribbon type) knot is invented in later centuries.
- Decision making: some parts (black dress's shape, hand holding the baby, etc.) must be invented. Considering historical context, physics, and reality, the conservator should make decisions about inpainting.

* Test panel (unknown Italian painting on wood panel)



* Geertgen tot Sint Jans_The Holy Kinship circa 1495



After over paint was removed



Before

treatment 1983



After

treatment 2021

* Geertgen tot Sint Jans_ Adoration of the Kings circa 1480–1485



Before

During cleaning

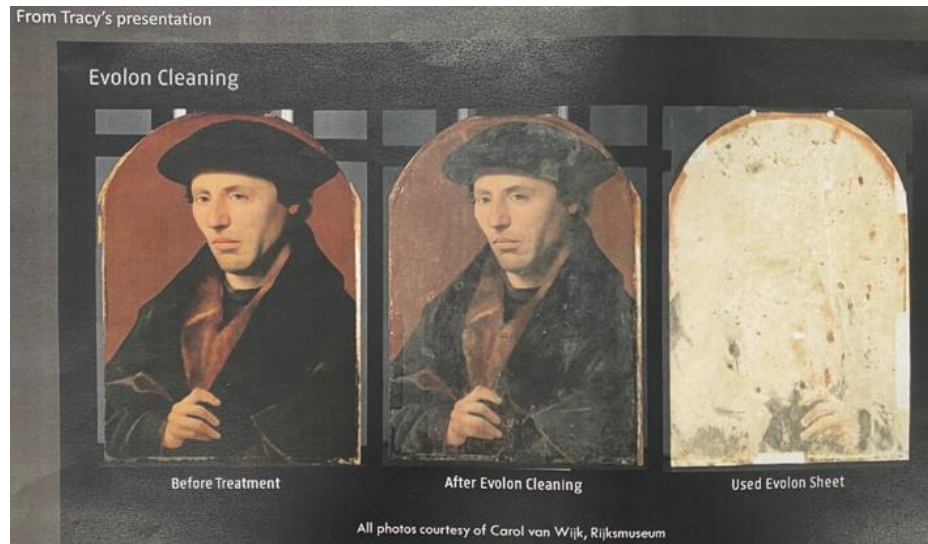


UV



Rock was invented
Based on the
historical context

*Removal of over painting: Evolon cleaning



Conclusion & Reflection

In our group, three people retouched the same painting in different ways. The damaged part of this painting mostly had a brown color, so it was appropriate to apply a neutral tone technique. At the same time, the vertical stripes of the original wood panel made it appropriate to apply *Tratteggio* to create a similar visual texture. On the contrary, it seems that imitative retouching needs to be applied under some limitations. The reason is that, for example, the hand holding the tool in the center of the painting is lost, so it is not possible to know the original drawing. If a conservator creates the drawing, the meaning of the painting could change. So, it was agreed that it was necessary to put only the color tone in this part and send the damage backward for the appreciation of the audience without irritation caused by the damage. After trying various techniques, I concluded that the most effective way to remove the distracting elements from the audience's appreciation is to use a mixture of techniques. I found that a decent retouching is to apply the appropriate technique differently for each damage, considering the surrounding color, ground texture, damage location, and semantic importance, to achieve wholeness.

Meanwhile, there were the following limitations in retouching practice: the actual size of the painting and the size of the paper print (A4) were different, so it was difficult to apply the retouching technique in detail. Since it was on the paper, watercolor stains remained. Moreover, the watercolor was in dried form; it was difficult to make enough paint of the same color as needed.

In my case, I thought about whether this retouching technique could be applied to contemporary artworks. For contemporary art, full reintegration of the surface often occurs, and it is common to use various materials, so there are limits to applying *Tratteggio* or pointillism to match colors. Also, loose pigments are often used with binders in the contemporary world, such as paraloid B67, I would improve my skills in using loose pigments.

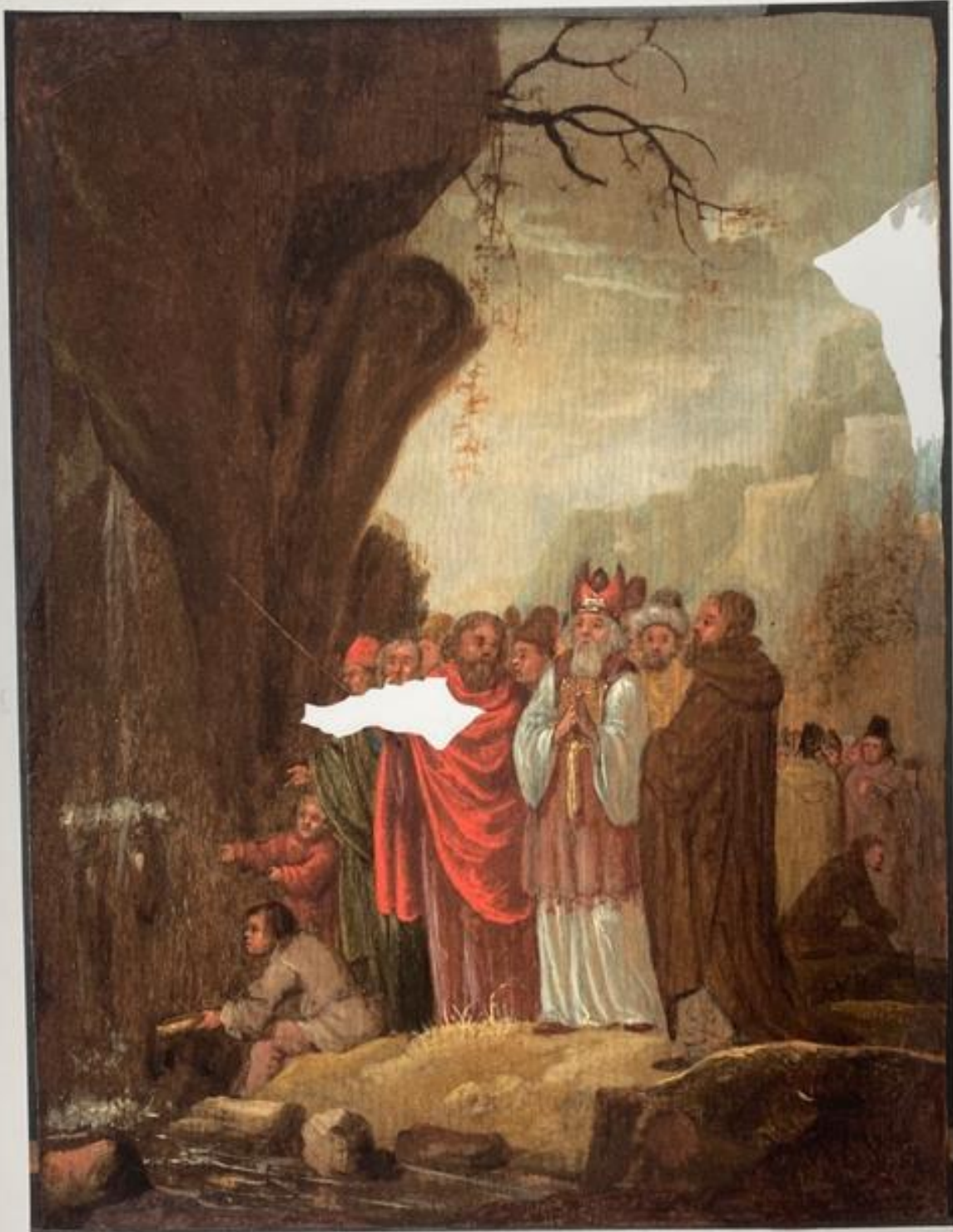
What was interesting from the guest lecture was that suggestions (inventions) of

original drawings at some levels were made in the actual conservation process. I learned that this should be decided based on knowledge from various fields, such as historical context and physics.

Appendix: results

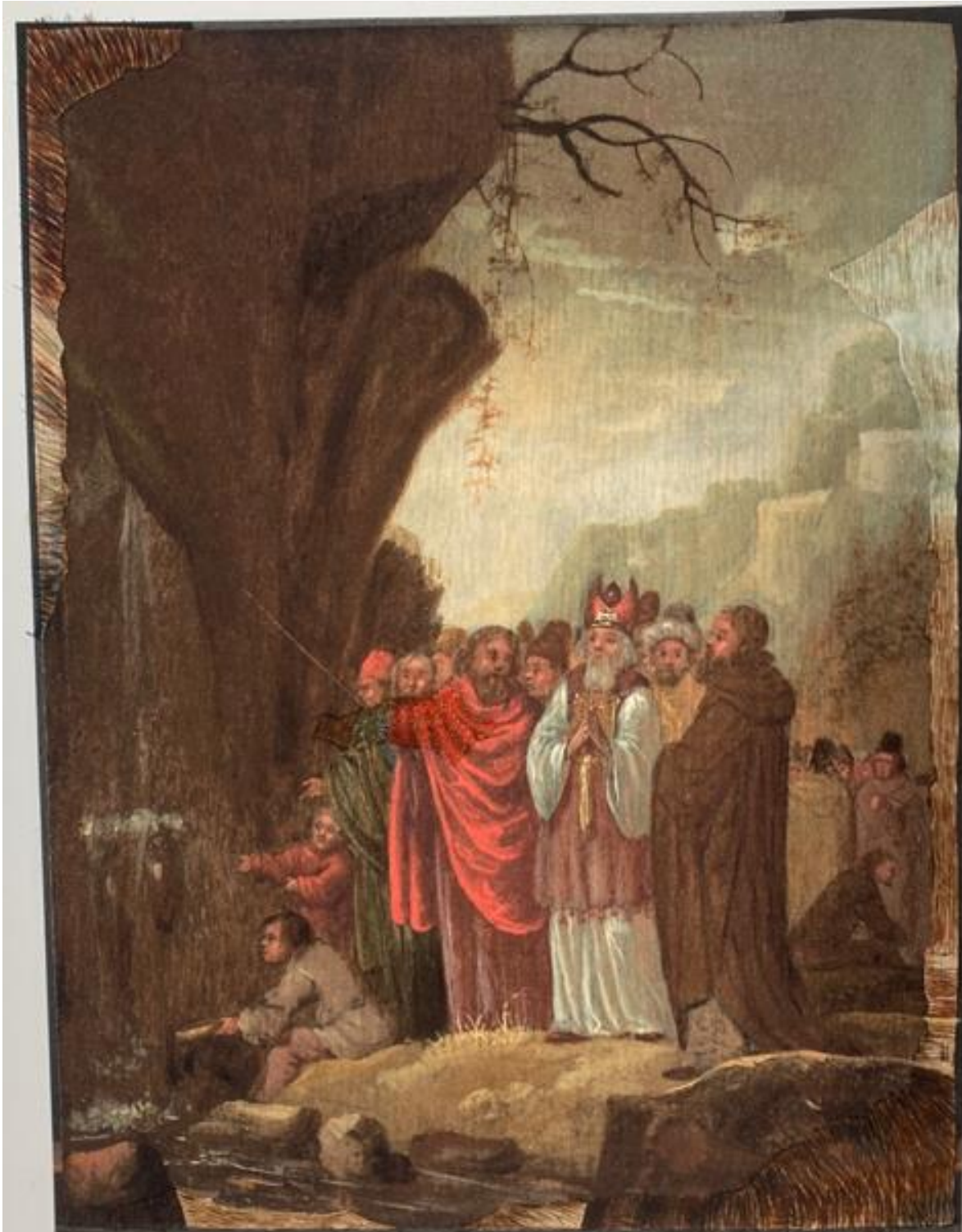


Fig1. The final result on a bigger scale



- Retouching by Kathleen Morgan, mostly used imitating retouching technique
- Imitative retouching

Fig 2. Other student's work (in the same group/ Kathleen Morgan from Painting)



- Retouching by Rosa Vos, mostly used Tratteggio technique
- Tratteggio

Fig 3. Other student's work (in the same group/ Rosa Vos from Painting)



Damaged tile / printed on paper



The best result – imitative retouching & neutral tone

Fig 4. A tile group's work: the best result



Fig 5. A tile group's work examples



Readings

Compulsory readings in preparation to the workshop

Brandi, Cesare. (1977) "Theory of Restoration, II." In *Historical and Philosophical Issues in Conservation of Cultural Heritage*, edited by Nicholas Stanley Price, Mansfield Kirby Talley Jr., Alessandra Melucco Vaccaro, 339-342. Los Angeles: The Getty Conservation Institute, 1996.

Betts, Pamela. (2011). Italian Differentiated Inpainting: A Review Inpainting. *Painting Conservation Catalog Volume III* (ed. Metzger, Catherine A). The Paintings Specialty Group of the American Institute for Conservation. 53-57

Mora, Paolo, Laura Mora and Paul Philippot. (1975) "Problems of Presentation." In *Historical and Philosophical Issues in Conservation of Cultural Heritage*, edited by Nicholas Stanley Price, Mansfield Kirby Talley Jr., Alessandra Melucco Vaccaro: 343-354. Los Angeles: The Getty Conservation Institute, 1996.

Muir, K. (2009). Approaches to the reintegration of paint loss: theory and practice in the conservation of easel paintings. *Studies in Conservation*, 54(sup1), 19–28. <https://doi.org/10.1179/sic.2009.54.Supplement-1.19> [Links to an external site.](#)

Wallert, A; G. Tauber, L. Murphy (2001). The Holy Kinship: A medieval masterpiece, Waanders publishers, Rijksmuseum, Amsterdam, 2001.

Complementary reading (focus on paintings conservation)

Bomford, David *Cleaning and Retouching: Changing Taste in the Restoration of Paintings; Restoration: is it acceptable?* British Museum Occasional Paper, no. 99, 1994: 33 – 40

Jessell, Bettina, "Helmut Ruhemann's Inpainting Techniques," in *JAIC* Vol 17 No 1, 1977: 1-8.

Melucco Vaccaro Alessandro, "Introduction to Part VI Reintegration of Losses", in ed. Nicholas Stanley Price, M. Kirby Talley Jr., Alessandro Melucco Vaccaro, in *Readings in Conservation: Historical and Philosophical Issues in Conservation of cultural Heritage*, 326-331 (Los Angeles, The Getty Conservation Institute, 1996).

Nadolny, Jilleen. "History of visual compensation for paintings." In *The Conservation of Easel Paintings*, edited by Joyce Hill Stoner and Rebecca Rushfield, 573-585. London: Routledge, 2012.

Philippot Albert and Paul Philippot, "The problem of the Integration of Lacunae in the Restoration of Paintings," in *Historical and Philosophical Issues in Conservation of cultural Heritage*, 335-338. First edition, 1959. Los Angeles, The Getty Conservation Institute, 1996.

Ramsay, L, "An evaluation of Italian Retouching Techniques," in *ABPR Conference 2000: Retouching and Filling* (2000) pp. 10 – 13.

Robinson, P. Light and Bright: "A Practical Approach to Restoration" in *The Picture Restorer* ABPR No. 3 pp. 5 – 8.

Ruheman, Helmut. (1968). Technique and Ethics of Retouching. *The Cleaning of Paintings*. (London; Faber and Faber). 269-279.

David Saunders, "Retouching: Colour vision and optical considerations," in *Retouching and Filling: Conference 2000 at the National Gallery London*. London, Association of British Picture Restorers, 2000.



Seymour, Kate and Giulia van den Burg.
<https://english.cultureelerfgoed.nl/publications/publications/2022/01/01/varnishing-and-inpainting>

Online sources

Workshop guide:

https://canvas.uva.nl/courses/48420/pages/workshop-guide-retouching-techniques-sbc4-emilie-froment-2024-2025?module_item_id=2131039 (Accessed on 20th. Oct.2024)

Retouching techniques:

<https://artenet.it/en/pictorial-inpainting-methods/>

<https://artenet.it/en/restoration-of-paintings/>

<https://artuk.org/discover/stories/sympathetic-retouching>

<http://www.toscanarestauroarte.it/ricopitt.asp?idMenu=8&idLingua=2>